

Python Functional Programming

First 20 Practice Questions — Full-Stack Aspirants

Questions only • No solutions

1. Square numbers using map()

Use map() to find the square of every number.

Input:

```
numbers = [2, 4, 6, 8, 10]
```

2. Find even numbers using filter()

Use filter() to find all even numbers.

Input:

```
numbers = [11, 24, 35, 42, 53, 66, 77, 80]
```

3. Find average using reduce()

Find the average of the numbers using functional programming.

Input:

```
numbers = [10, 20, 30, 40, 50]
```

4. Average student marks

Calculate the average marks of all students.

Input:

```
marks = [75, 82, 68, 90, 85, 72]
```

5. Average student age

Find the average age of the students.

Input:

```
students = [  
    {"name": "Anu", "age": 20},  
    {"name": "Rahul", "age": 22},  
    {"name": "Meera", "age": 21},  
    {"name": "Arun", "age": 23}  
]
```

6. Average employee salary

Calculate the average salary of all employees.

Input:

```
employees = [  
    {"name": "Anu", "salary": 30000},  
    {"name": "Rahul", "salary": 45000},  
    {"name": "Meera", "salary": 55000},  
    {"name": "Arun", "salary": 40000}  
]
```

7. Average salary above ■30,000

Find the average salary of employees whose salary is greater than ■30,000.

Input:

```
employees = [  
  {"name": "Anu", "salary": 25000},  
  {"name": "Rahul", "salary": 45000},  
  {"name": "Meera", "salary": 55000},  
  {"name": "Arun", "salary": 28000},  
  {"name": "Vijay", "salary": 60000}  
]
```

8. Sum, count and average

Using functional programming, find the sum, count, and average of the numbers.

Input:

```
numbers = [15, 25, 35, 45, 55, 65]
```

9. Average product price

Calculate the average price of all products.

Input:

```
products = [  
  {"name": "Laptop", "price": 55000},  
  {"name": "Mouse", "price": 1200},  
  {"name": "Keyboard", "price": 2500},  
  {"name": "Monitor", "price": 15000}  
]
```

10. Average order amount

Calculate the average order amount.

Input:

```
orders = [  
  {"id": 101, "amount": 1500},  
  {"id": 102, "amount": 2500},  
  {"id": 103, "amount": 3500},  
  {"id": 104, "amount": 2000}  
]
```

11. Average completed order amount

Calculate the average amount of only completed orders.

Input:

```
orders = [  
  {"id": 101, "amount": 1500, "status": "completed"},  
  {"id": 102, "amount": 2500, "status": "pending"},  
  {"id": 103, "amount": 3500, "status": "completed"},  
  {"id": 104, "amount": 2000, "status": "cancelled"},  
  {"id": 105, "amount": 4000, "status": "completed"}  
]
```

12. Students above average

Calculate the average mark and find students who scored above the average.

Input:

```
students = [  
  {"name": "Anu", "marks": 75},  
  {"name": "Rahul", "marks": 55},  
  {"name": "Meera", "marks": 90},  
  {"name": "Arun", "marks": 65},  
  {"name": "Vijay", "marks": 85}  
]
```

13. Employees above average salary

Calculate the average salary and find employees whose salary is above the average.

Input:

```
employees = [  
  {"name": "Anu", "salary": 30000},  
  {"name": "Rahul", "salary": 50000},  
  {"name": "Meera", "salary": 40000},  
  {"name": "Arun", "salary": 70000},  
  {"name": "Vijay", "salary": 35000}  
]
```

14. Products below average price

Calculate the average product price and find products cheaper than the average.

Input:

```
products = [  
  {"name": "Laptop", "price": 60000},  
  {"name": "Mouse", "price": 1000},  
  {"name": "Keyboard", "price": 2500},  
  {"name": "Monitor", "price": 18000},  
  {"name": "Phone", "price": 40000}  
]
```

15. Numbers greater than average

Find all numbers greater than the average.

Input:

```
numbers = [10, 25, 30, 45, 50, 60, 75]
```

16. Average temperature

Calculate the average temperature and find temperatures above average.

Input:

```
temperatures = [28, 31, 29, 35, 32, 27, 30]
```

17. Average customer spending

Calculate the average amount spent by customers.

Input:

```
customers = [  
  {"name": "Anu", "spent": 5000},  
  {"name": "Rahul", "spent": 3500},  
  {"name": "Meera", "spent": 7500},  
  {"name": "Arun", "spent": 4000}  
]
```

18. Average age of active users

Find the average age of active users only.

Input:

```
users = [  
  {"name": "Anu", "age": 22, "active": True},  
  {"name": "Rahul", "age": 25, "active": False},  
  {"name": "Meera", "age": 28, "active": True},  
  {"name": "Arun", "age": 24, "active": True},  
  {"name": "Vijay", "age": 30, "active": False}  
]
```

19. Average successful transaction

Find the average amount of successful transactions.

Input:

```
transactions = [  
  {"id": 1, "amount": 1000, "status": "success"},  
  {"id": 2, "amount": 2500, "status": "failed"},  
  {"id": 3, "amount": 3000, "status": "success"},  
  {"id": 4, "amount": 1500, "status": "success"},  
  {"id": 5, "amount": 4000, "status": "failed"}  
]
```

20. Average monthly sales

Calculate the average monthly sales.

Input:

```
sales = [  
  {"month": "January", "amount": 50000},  
  {"month": "February", "amount": 65000},  
  {"month": "March", "amount": 55000},  
  {"month": "April", "amount": 70000},  
  {"month": "May", "amount": 60000}  
]
```